

Appendix P4B: Pipe Material Selection Guide

- Instruction:
1. This shall serve as a guide to the Tenderers. Final pipe material, lining and joint type selection shall be approved by the S.O. before installation.
 2. There can be more joint types than listed below with advance in technology and with respect to special pipe fittings types. All subject to approval by S.O.

Fluid Code	Services	Size (mm) Note 1	Material 1 Note 2	Material 2 Note 2	Material 3 Note 2	Internal Lining Note 3	Installation Note 4	Joint Type Note 5	Remarks Note 6	Pressure Rating	Colour based on BS 5252 / BS 4800: 2011 (Accurate RGB)	Colour Code Name	Colour Code
AA	Aeration Air	All	SS16			-	EXP, EMB	F,W,S		< 10 Bar		Light Blue	18 E 51
AA	Aeration Air	All	CPVC			-	SUB	G,S,F,W		< 10 Bar		Light Blue	18 E 51
AHP	Air High Pressure	All	SS16			-	EXP, EMB, SUB	F,W, S	Barrier protection provided to hot pipes where potential for accidental touch	>= 10 Bar		Light Blue	20 E 51
ALP	Air Low Pressure	All	SS16			-	EXP, EMB, SUB	F,W,S	Barrier protection provided to hot pipes where potential for accidental touch	< 10 Bar		Blue	18 E 49
NHS	Ammonium Sulphate	All	CPVC			-	EXP, EMB, SUB	G,S,F,W				Lettuce Green	12 E 55
ASC	Antiscalant	All	CPVC			-	EXP, EMB, SUB	G,S,F,W				Jade	16 D 45
BL	BTF Liquor	All	FRP			-	EXP, EMB, SUB	F, SOC, W	Specific FRP corrosion resistant liner (eg. epoxy, vinyl ester, etc.) to be confirmed with manufacturer/supplier			Dark Green	14 D 45
BL	BTF Liquor	All	CPVC	FRP		-	EXP, EMB, SUB	G,S,F,W, SOC (applicable for FRP)	Specific FRP corrosion resistant liner (eg. epoxy, vinyl ester, etc.) to be confirmed with manufacturer/supplier			Dark Green	14 D 45
CDG	Carbon Dioxide (Gas)	All	SS16			-	EXP, EMB	W, F, S				Grey	00 A 07
CHWR	Chilled Water Return	All	CS			-	EXP, EMB	F, W,S				Island Green	12 D 43
CHWS	Chilled Water Supply	All	CS			-	EXP, EMB	F, W,S				Island Green	12 D 43
CIPR	CIP Return	All	FRP			-	All	F, SOC, W				Light Blue	20 E 50
CIPS	CIP Supply	All	FRP			-	All	F, SOC, W				Light Blue	20 E 50
CIT	Citric Acid	All	CPVC			-	EXP, EMB, SUB	G,S,F,W				Orange	06 E 55
SC	Condensate	All	UPVC	PVDF	CS	PTFE liner for CS option	EXP, EMB	G,S,F,W	UPVC should not be used if the condensate has the possibility to be high temperature, > 60 degrees C. PVDF option is suitable up to 120 degrees C. CS shall be used for condensate from steam pipework.			Dark Green	14 D 45
DWAS	Dewatered Waste Activated Sludge	All	SS16				EXP, EMB, SUB	W, F				Brown	08 D 45
DC	Dewatering Centrate	All	CSEL	CSPL		Fusion bonded epoxy lining or 100% solids PU lining	All	W, F	Pipework to be designed with well-planned flanged connections to facilitate opening for removal of potential struvite build-up.			Dark Green	14 D 45
DCW	Dewatering Centrate (WAS Pre-dewatering)	All	CSEL	CSPL		Fusion bonded epoxy lining or 100% solids PU lining	All	W, F	Pipework to be designed with well-planned flanged connections to facilitate opening for removal of potential struvite build-up.			Dark Green	14 D 45

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DS	Digested Sludge	All	SS16			-	EXP, EMB, SUB	S, W, F				Pale Violet Red	0
DUW	Domestic Used Water	All	CSCL			CL	EXP, EMB	W, F				Green	14 E 53
DUW	Domestic Used Water	All	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC	Flange to be avoided as far as possible.			Green	14 E 53
FC	Ferric Chloride	All	CPVC	PP	PVDF	-	EXP, EMB, SUB	W, F, G, S, FB (applicable for PVDF)	Double containment having see through pipe section at low points, see through catchpots and leak detection			Light Green	14 C 31
FPW	Fire Water	All	SS04	SS16		-	EXP, EMB	W,F,S	SS316 (i.e. SS16) material to be used where location of pipework is assessed to be of higher corrosion potential.			Red	04 E 55
FPW	Fire Water	All	DICL			CL	BUR	F, SOC				Red	04 E 55
FW	Food Waste	All	SS16			-	EXP, EMB, SUB	W,F,S				Black	00 E 53
FD	Foul (Sanitary) Drain	All	UPVC			-	EXP, EMB	G,S,F,W				Black	00 E 53
FD	Foul (Sanitary) Drain	All	DIPU	DIEL	VCP	100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC	For underground pipework and with haunching where applicable DICL pipe for Foul (Sanitary) Drain in CWRP noted to be corroded. So avoid DICL.Flange to be avoided as far as possible.			Black	00 E 53
FA	Foul Air	All	FRP	SS16		-	All	F, SOC, W	SS16 pipework shall be used for Foul Air at elevated temperatures.			White	00 E 55
FOR	Fuel Oil (Return)	All	CS			-	EXP, EMB, SUB	W,F,S				Dark Purple	24 C 37
FOS	Fuel Oil (Supply)	All	CS			-	EXP, EMB, SUB	W,F,S				Dark Purple	24 C 37
GB	Gas (Biogas)	All	SS16			-	EXP, EMB, SUB	S, W, F				Canary Yellow with Red Banding	10 E 55
GB	Gas (Biogas)	All	SS16			-	EXP, EMB, SUB	S, W, F				Canary Yellow with Red Banding	10 E 55
GW	Greasy Waste	All	SS16	CSEL		-	EXP, EMB, SUB	W,F,S				Black	00 E 53
GR	Grit	All	HDPE			-	EXP, EMB	F, SOC, W	Pipework shall be provided with long radius bends and include provision for jetting to clear blockages.			Black	00 E 53
HPHW	High Pressure Hot Water	All	SS16			-	EXP, EMB, SUB	W, F	Applications for non-heat exchange hot water pipework only. With jacket insulation conforming to code.			Cobalt Blue with Red Banding	18 E 53
HWR	Hot Water (Return)	All	SS16			-	EXP, EMB, SUB	W, F				Yellow Red	06 E 51
HWS	Hot Water (Supply)	All	SS16			-	EXP, EMB, SUB	W, F				Yellow Red	06 E 51

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Fluid Code	Services	Size (mm) Note 1	Material 1 Note 2	Material 2 Note 2	Material 3 Note 2	Internal Lining Note 3	Installation Note 4	Joint Type Note 5	Remarks Note 6	Pressure Rating	Colour based on BS 5252 / BS 4800: 2011 (Accurate RGB)	Colour Code Name	Colour Code
HS	Hydrolysed Sludge	All	SS16				EXP, EMB, SUB	W, F				Moss Green	14 E 56
IUW	Industrial Used Water	All	CSPL	CSEL		100% solids PU lining or Fusion Bonded Epoxy lining	EXP, EMB	W, F				Green	14 E 53
IUW	Industrial Used Water	All	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC	Flange to be avoided as far as possible.			Green	14 E 53
IW	Industrial Water	All	CSEL	CSPL	FRP	Fusion bonded epoxy lining or 100% solids PU lining	EXP, EMB	W, F				Yellow	08 E 51
IW	Industrial Water	All	DIPU	DIEL	FRP	100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC	Flange connection to be avoided as far as possible.			Yellow	08 E 51
IA	Instrument Air	All	SS16			-	EXP, EMB, SUB	F,W,S		< 10 Bar		Light Blue	18 E 51
LH	Lime Hydrated	All	CS	CPVC		-	EXP, EMB	W, F	G,S,F,W joints suitable is using CPVC pipework.			Light Green	14 E 50
LSL	Lime Sludge	All	HDPE			-	EXP, EMB, SUB	F, SOC, W	Long radius beds shall be used for lime slude pipework.			Light Green	14 E 50
LS	Lime Solution	All	HDPE			-	EXP, EMB, SUB	S,F,W	When LS is in a slurry form only swept bends or long radius bends shall be used.			Light Green	14 E 50
LS	Lime Solution	All	EPDM	NBR	CR	-	EXP	F (with crimping as necessary)	These are non metallic flexible hoses as necessitated by maintenance requirements.			Light Green	14 E 50
LPHW	Low Pressure Hot Water	All	SS16			-	EXP, EMB, SUB	W, F	Applications for non-heat exchange hot water pipework only. With jacket insulation conforming to code.			Mace	06 D 45
MF	Membrane Filtrate	All	SS16			-	EXP, EMB, SUB	F, SOC, W				Light Blue	20 E 50
MF	Membrane Filtrate	All	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC, W				Light Blue	20 E 50
NW	NEWater	All	CSEL	SS16	CSPL	Fusion bonded epoxy lining or 100% solids PU lining	EXP, EMB	W, F	From common discharge header of UV Modules and downstream pipework.			Purple	22 C 37
NW	NEWater	All	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC, W	Flange connection to be avoided as far as possible.			Purple	22 C 37
PD	Plant Drain	<=100	SS16	UPVC	CPVC	-	EXP, EMB	F, S				Dark Green	14 D 45
PD	Plant Drain	>= 100	FRP	SS16		-	EXP, EMB, SUB	F, SOC, W				Dark Green	14 D 45
PD	Plant Drain	>= 100 to < 1000	VCP			-	EMB	SOC	For non process applications			Dark Green	14 D 45
PD	Plant Drain	>= 100 to < 1000	DIPU	DIEL	CPVC	100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC	For process applications, Flange connection to be avoided as far as possible. CPVC shall be used where acidic process fluids, such as biogas condensate, will be present.			Dark Green	14 D 45
PD	Plant Drain	>= 100 to < 1000	HDPE			-	EXP, BUR	F, SOC, W	For plant drain flows with heavy grit content, such as at the headworks.			Dark Green	14 D 45

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PD	Plant Drain	>= 1000	RCP			HDPE	BUR	SOC	Only for underground pipework (eg. for conveying plant process overflows). Lining to comply with the requirements specified in G3.41 – Material specification for sewer pipes.			Dark Green	14 D 45
PLE	Plant Effluent	<= 80	SS16			-	EXP, EMB, SUB	S, W, F				Cobalt Blue	18 E 53
PLE	Plant Effluent	>= 100	SS16			-	EXP, EMB, SUB	W, F				Cobalt Blue	18 E 53
PLE	Plant Effluent	>=100	DIPU	DIEL	FRP	100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC, W				Cobalt Blue	18 E 53
G1	Plant Water (ROP)	All	FRP	DIPU			BUR	F, SOC, W				Light Green	14 E 50
G1	Plant Water (ROP)	All	SS16	CSEL			EXP, EMB	W, F				Light Green	14 E 50
G2D	Plant Water (DLM MF)	<= 80	SS16			-	EXP, EMB, SUB	S, W, F		< 10 Bar		Cobalt Blue	18 E 53
G2D	Plant Water (DLM MF)	>= 100	SS16			-	EXP, EMB, SUB	W, F		< 10 Bar		Cobalt Blue	18 E 53
G2D	Plant Water (DLM MF)	>=100	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC, W	Flange connection to be avoided as far as possible.	< 10 Bar		Cobalt Blue	18 E 53
G2I	Plant Water (ILM MF)	<= 80	SS16			-	EXP, EMB, SUB	S, W, F		< 10 Bar		Cobalt Blue with black banding	18 E 53
G2I	Plant Water (ILM MF)	>= 100	SS16			-	EXP, EMB, SUB	W, F		< 10 Bar		Cobalt Blue with black banding	18 E 53
G2I	Plant Water (ILM MF)	>=100	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC, W	Flange connection to be avoided as far as possible.	< 10 Bar		Cobalt Blue with black banding	18 E 53
G2HP	Plant Water High Pressure	<= 80	SS16			-	EXP, EMB, SUB	S, W, F		>= 10 Bar		Cobalt Blue	18 E 53
G2HP	Plant Water High Pressure	>= 100	SS16			-	EXP, EMB, SUB	W, F		>= 10 Bar		Cobalt Blue	18 E 53
G2HP	Plant Water High Pressure	>=100	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC, W	Flange connection to be avoided as far as possible.	>= 10 Bar		Cobalt Blue	18 E 53
PO	Polyelectrolyte	All	CPVC			-	EXP, EMB, SUB	G,S,F,W	Use of flanged pipework is recommended. This will enable the pipework to be dismantled in the event of a blockage.			Grey	00 A 07
PE	Primary Effluent	All	SS16			-	EXP, EMB, SUB	W, F				Green	14 E 53
PE	Primary Effluent	>=100	DIPU	DIEL	FRP	100% solids PU lining or Fusion Bonded Epoxy lining for the DI pipe material	BUR	F, SOC, W				Green	14 E 53
PE	Primary Effluent	All	HDPE			-	EXP, EMB, SUB	SOC, W	Grit containing fluid			Green	14 E 53

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PI	Primary Influent	All	SS16			-	EXP, EMB, SUB	W, F				Green	14 E 53
PI	Primary Influent	All	DIPU			100% solids PU lining for the DI pipe material	BUR	F, SOC	Flange connection to be avoided as far as possible.			Green	14 E 53
PSM	Primary Scum	All	SS16			-	EXP, EMB, SUB	W, F				Yellow Red	08 D 41
PSM	Primary Scum	All	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining	BUR	F, SOC	For underground pipework of this application.			Yellow Red	08 D 41
PSD	Primary Sludge Domestic	All	SS16				EXP, EMB, SUB	W, F				Black	00 E 53
PSD	Primary Sludge Domestic	All	FRP				BUR	F, SOC, W	For underground pipework of this application. DICL pipe for Primary Sludge in CWRP noted to be corroded. So avoid DICL pipe. Flange to be avoided as far as possible.			Black	00 E 53
PSI	Primary Sludge Industrial	All	SS16				EXP, EMB, SUB	W, F				Black	00 E 53
PSI	Primary Sludge Industrial	All	FRP				BUR	F, SOC, W	For underground pipework of this application. DICL pipe for Primary Sludge in CWRP noted to be corroded. So avoid DICL pipe. Flange to be avoided as far as possible.			Black	00 E 53
RAS	Return Activated Sludge	All	SS16				EXP, EMB, SUB	W, F				Brown	08 D 45
RAS	Return Activated Sludge	All	DICL			CL	BUR	F, SOC	For underground pipework of this application. Flange to be avoided as far as possible.			Brown	08 D 45
ROC	RO Concentrate	All	SS16				EXP, EMB, BUR	W, F	RO Modules concentrate outlet up to and including header and tee fittings.			Cobalt Blue	18 E 53
ROC	RO Concentrate	All	FRP				EXP, EMB, BUR	F, SOC, W	RO Concentrate pipework downstream of discharge header.			Cobalt Blue	18 E 53
ROF	RO Feed	All	SS16				All	W, F	For feed to RO Modules including pipework between 1st and 2nd Pass RO Modules			Light Blue	20 E 50
ROP	RO Permeate	All	FRP	DIPU			BUR	F, SOC, W	RO Permeate from 2nd Pass RO up to and including common discharge header and header tees.			Light Green	14 E 50
ROP	RO Permeate	All	SS16	CSEL			EXP, EMB	W, F	RO Permeate from 2nd Pass RO up to and including common discharge header and header tees. Individual UV Modules discharge pipework.			Light Green	14 E 50
S	Sample	All	CPVC	SS16			EXP, EMB	S	If at any point the sample line exceeds 0.5 bar the entire pipeline shall be stainless steel. Stainless steel shall be used for any sample lines >80 degrees C			White	00 E 55
AS	Scour Air	All	SS16				EXP, EMB, SUB	F,W,S		< 10 Bar		Light Blue	18 E 51
SG	Screenings	All	SS16				EXP, EMB, SUB	W, F				Yellow Red	08 D 41
SSM	Secondary Scum	All	DIPU	DIEL		100% solids PU lining or Fusion Bonded Epoxy lining	BUR	F, SOC	For underground pipework of this application.			Yellow Red	08 D 41

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SSM	Secondary Scum	All	SS16				EXP, EMB, SUB	W, F				Yellow Red	08 D 41
SBS	Sodium Bisulphite	All	CPVC				EXP, EMB, SUB	W, F, G, S				Violet	22 B 27
SHYD	Sodium Hydroxide	All	CPVC				EXP, EMB, SUB	W, F, G, S	Double containment having see through pipe section at low points, see through catchpots and leak detection			Purple	24 E 53
SHYP	Sodium Hypochlorite	All	CPVC				EXP, EMB, SUB	W, F, G, S,	Double containment having see through pipe section at low points, see through catchpots and leak detection			Purple	24 E 53
STHP	Steam High Pressure	All	SS16				EXP, EMB, SUB	W, F				Dawn Grey	10 A 03
STLP	Steam Low Pressure	All	SS16				EXP, EMB, SUB	W, F				Dawn Grey	10 A 03
SA	Sulphuric Acid	All	PVDF				EXP, EMB, SUB	F, FB	Non hot chemical and all concentrations; Double containment having see through pipe section at low points, see through catchpots and leak detection			Purple Blue	20 C 33
TCD	Thickening Centrate Domestic	All	CSEL	CSPL		Fusion bonded epoxy lining or 100% solids PU lining	All	W, F				Dark Green	14 D 45
TCI	Thickening Centrate Industrial	All	CSEL	CSPL		Fusion bonded epoxy lining or 100% solids PU lining	All	W, F				Dark Green	14 D 45
TA	Treated Air	All	FRP			-	All	F, SOC, W				Blue	20 E 56
WAS	Waste Activated Sludge	All	DICL			CL	BUR	F, SOC	For underground pipework of this application.			Brown	08 D 45
WAS	Waste Activated Sludge	All	SS16				EXP, EMB, SUB	W, F				Brown	08 D 45
WASD	Waste Activated Sludge Domestic	All	DICL			CL	BUR	F, SOC	For underground pipework of this application.			Brown	08 D 45
WASD	Waste Activated Sludge Domestic	All	SS16				EXP, EMB, SUB	W, F				Brown	08 D 45
WASI	Waste Activated Sludge Industrial	All	DICL			CL	BUR	F, SOC	For underground pipework of this application.			Brown	08 D 45
WASI	Waste Activated Sludge Industrial	All	SS16			-	EXP, EMB, SUB	W, F				Brown	08 D 45
WC	Waste Chemical	All	CPVC			-	EXP, EMB, SUB	G,S,F,W	Double containment having see through pipe section at low points, see through catchpots and leak detection			Black	00 E 53
TPS	Thickened Primary Sludge	All	SS16			-	EXP, EMB, SUB	W, F,				Black	00 E 53
TPS	Thickened Primary Sludge	All	FRP			-	BUR	F, SOC, W	For underground pipework of this application. DICL pipe for Primary Sludge in CWRP noted to be corroded. So avoid DICL pipe. Flange to be avoided as far as possible.			Black	00 E 53

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FH	Fire Hydrant Water	All	SS04	SS16		-	EXP, EMB, SUB	W, F	SS316 to be used as the material in areas with relatively higher corrosion potential (eg. Sludge Building)			Red	04 E 53
FH	Fire Hydrant Water	All	DICL			CL	BUR	F, SOC				Red	04 E 53
OGW	Organic Waste	All	SS16	CSEL		-	EXP, EMB, SUB	W,F,S				Brown	06 C 40
PW	Potable Water	All	SS16			-	EXP, EMB, SUB	W, F	Specified by CP 48: 2005 Code of Practice for Water Services Article 5.1.5			Ribbon Blue/Cirrus	18 E 50
PW	Potable Water	All	DICL			CL	BUR	F, SOC	Specified by CP 48: 2005 Code of Practice for Water Services Article 2.2.5			Ribbon Blue/Cirrus	18 E 50
WWS	Wet Weather Sludge	All	SS16			-	EXP, EMB, SUB	W, F				Brown	08 C 40
WWS	Wet Weather Sludge	All	FRP			-	BUR	F, SOC, W	For underground pipework of this application. DICL pipe for Primary Sludge in CWRP noted to be corroded. So avoid DICL pipe. Flange to be avoided as far as possible.			Brown	08 C 40

- Notes:**
- 1. Inequality symbols**

">" Greater Than

"<" Less Than

"<=" Less Than or Equal To

">=" Greater Than or Equal To

"All" All sizes

2. Abbreviations for piping materials as follows:

Refer to Process Legends Drawings

3. - : Bare i.e. without lining

4. Installation configuration:

EXP: Exposed (interior or exterior)

BUR: Buried

EMB: Embedded Or Encased (in concrete)

SUB: Submerged

All: All Installations

5. F: Flanged of same PN rating as the pipe

G: Solvent Cement Glue

SOC: Socket and Spigot

W: Welded (Butt) / Lamination for FRP pipes / PVC welding for uPVC and cPVC pipes / Thermofusion welding for HDPE or PP pipes / Brazing for Cu pipes

S: Screwed (Threaded) / Compression type fittings for Cu pipes

FB: Fusion Bonded

6. Insulated pipe shall be prime coated only.

7. All piping pressure ratings to be confirmed by Contractor for S.O. review

8. Refer to General and Particular Specifications for pipe material details
For information on colour code, refer British Standard BS 4800/5252

9. Visit website https://www.e-paint.co.uk/bs5252_colour_chart.asp and click on the specific colour to reveal other details such as the RGB nos.

10 A restrained mechanical coupling incorporating a grooved joint can be used as an alternative to welding on long straight, above ground, pipe runs. This applies to CS, CSEL, CSPL, SS04 and SS16 pipelines.

11 The Contractor shall provide the pipe material shown in the Drawings.
The alternative materials listed above may be used, subject to S.O. approval.